TIØ4145 Corporate Finance Chapters 14 - 15

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Capital Structure

- **Today**
  - **Chapter 14:** Capital Structure in a Perfect Market
  - **Chapter 15:** Debt and Taxes
- **Next week**
  - **Chapter 16:** Financial Distress, Managerial Incentives, and Information
  - **Chapter 17:** Payout Policy

- *Perfect capital markets:*
  - Investors and firms can trade the same set of securities at competitive market prices equal to the present value of their future cash flows.
  - There are no taxes, transaction costs, or issuance costs associated with security trading.
  - A firm's financing decisions do not change the cash flows generated by its investments, nor do they reveal new information about them.

Learning objectives

- Describe the capital structure that the firm should choose.
- List the three conditions that make capital markets perfect.
- Discuss the implications of MM Proposition I, and the roles of homemade leverage and the Law of One Price in the development of the proposition.
- Calculate the cost of capital for levered equity according to MM Proposition II.
- Illustrate the effect of a change in debt on weighted average cost of capital in perfect capital markets.
- Calculate the market risk of a firm's assets using its unlevered beta.
- Illustrate the effect of increased leverage on the beta of a firm's equity.
- Compute a firm's net debt.
- Discuss the effect of leverage on a firm's expected earnings per share.
- Explain why perfect capital markets neither create nor destroy value.
- Calculate the interest tax shield, given the corporate tax rate and interest payments.
- Calculate the value of a levered firm.
- Calculate the weighted average cost of capital with corporate taxes.
- Describe the effect of personal taxes on the corporate tax benefits of leverage.

Capital Structure in Perfect Capital Markets without Corporate Taxation

- Capital structure describes the composition of financing:
  - Equity financing (limited vs. unlimited liability, control rights vs. no control rights, preferred vs. common stock, different tax rules)
  - Bank Loans (secured vs. unsecured, short vs. long term)
  - Leasing
  - Factoring (selling accounts receivables)
  - Convertible bonds (hybrid of equity and debt financing)
  - many more.
- In this lecture we assume: **Plain Equity** vs. **Plain Debt** in the following sense: debt holders will be satisfied first and equity holders receive whatever is left.

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Key assumptions

- Both investors and firms have equal access to the same competitive capital markets.
- No market entry or exit barriers
- No personal preferences concerning trading location and partner
- Perfect information (prices are assumed to be known to all consumers and producers)
- No costs for transaction, issuance, entry, information, trading, etc.
- A firm's financing decisions does not change the cash flows generated by its investments, nor do they reveal new information about them.

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Example

- Assume a firm with the following balance sheet (book values):

Assets		Liabilities & Equity	
Bank Account	800	Equity	800

- Imagine that the firm faces the following investment opportunity: For an initial investment of **800 in *t=0***, the project will generate cash flows of either **1400 (strong economy)** or **900 (weak economy) in *t=1***. Both scenarios are **equally likely.**
- Assume that **the risk free rate is 5%,** the **market risk premium is 6%**, and the **beta** of this investment is **5/3**=1.667.
- What is the required return for this investment and what is its NPV?
- What happens to the value of the firm if the project will be accepted?

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Example

	Unlevered Firm = Unlevered Equity	
Date 0 Initial value		
Date 1 – Strong Cash Flow/Return		
Date 1 – Weak Cash Flow /Return		
Date 1 – Expected Cash Flow /Return		
	Cash flow	Return

Assets		Liabilities & Equity	
Bank Account	800	Equity	800

- Initial investment of 800 in *t=0*
- Cash flows: either 1400 or 900 in *t=1*. Both are equally likely.
- Risk free rate is 5%, the market risk premium is 6%, and the beta is 5/3=1.667.

- What is the required return for this investment and what is its NPV?

- What happens to the value of the firm if the project will be accepted? Answer by filling in the table.

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Financing a Firm with Equity

- The project cash flows depend on the overall economy and thus contain market risk. As a result, you demand a risk premium over the current risk-free interest rate of 5% to invest in this project.
  - Cash flow: $1400*0.5+$900*0.5=$1150
  - CAPM: 5%+5/3*6%=15%
- What is the NPV of this investment opportunity?

$$NPV = -\$800 + \frac{\$1150}{1.15} = -\$800 + \$1000 = \$200$$

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Financing a Firm with Equity

- If you finance this project using only equity, how much would investors be willing to pay for the firm's shares?

$$PV(\text{equity cash flows}) = \frac{\$1150}{1.15} = \$1000$$

  - If you can raise $1000 by selling equity in the firm, after paying the investment cost of $800, you can keep the remaining $200, the NPV of the project NPV, as a profit.

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Financing a Firm with Equity

- Unlevered Equity
  - Equity in a firm with no debt
- Because there is no debt, the cash flows of the unlevered equity are equal to those of the project.

Example

	Unlevered Firm = Unlevered Equity		*Levered Equity*		*Debt*	
Date 0 Initial value	1000					
Date 1 – Strong Cash Flow/Return	1400					
Date 1 – Weak Cash Flow /Return	900					
Date 1 – Expected Cash Flow /Return	1150					

Table 1

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Financing a Firm with Equity

- **Shareholder's returns are either 40% or –10%.**
  - The expected return on the unlevered equity is:
    - ½ (40%) + ½(–10%) = 15%.
    - The cost of capital of the project is 15%,
    - Because the risk of unlevered equity equals the risk of the project, shareholders are earning an appropriate return for the risk they are taking.

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Example

- Suppose you decide to borrow $500 initially, fill in **Debt** column.

	Unlevered Firm = Unlevered Equity		*Levered Equity*		*Debt*	
Date 0 Initial value	1000					
Date 1 – Strong Cash Flow/Return	1400	40%				
Date 1 – Weak Cash Flow /Return	900	-10%				
Date 1 – Expected Cash Flow /Return	1150	15%				

Table 1

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Financing a Firm with Debt and Equity

- Suppose you decide to borrow $500 initially, in addition to selling equity.
  - Because the project's cash flow will always be enough to repay the debt, the debt is risk free and you can borrow at the risk-free interest rate of 5%. You will owe the debt holders:
    - $500 × 1.05 = $525 in one year.

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Example

	Unlevered Firm = Unlevered Equity		*Levered Equity*		*Debt*	
Date 0 Initial value	1000				500	
Date 1 – Strong Cash Flow/Return	1400	40%			525	5%
Date 1 – Weak Cash Flow /Return	900	-10%			525	5%
Date 1 – Expected Cash Flow /Return	1150	15%			525	5%

Table 1

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Financing a Firm with Debt and Equity

- Levered Equity
  - Equity in a firm that also has debt outstanding
- Given the firm's $525 debt obligation, your shareholders will receive only $875 ($1400 – $525 = $875) if the economy is strong and $375 ($900 – $525 = $375) if the economy is weak.

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Example

	Unlevered Firm = Unlevered Equity		*Levered Equity*		*Debt*	
Date 0 Initial value	1000				500	
Date 1 – Strong Cash Flow/Return	1400	40%	875		525	5%
Date 1 – Weak Cash Flow /Return	900	-10%	375		525	5%
Date 1 – Expected Cash Flow /Return	1150	15%	625		525	5%

Table 1

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Financing a Firm with Debt and Equity

- What price *E* should the levered equity sell for?
- Which is the best capital structure choice for the entrepreneur?

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Modigliani and Miller's Proposition I:

- *In a perfect capital market without corporate taxation, the total value of a firm is equal to the market value of the total cash flows generated by its assets and is not affected by its choice of capital structure.*

The American Economic Review

VOLUME XLVIII JUNE 1958 NUMBER THREE

THE COST OF CAPITAL, CORPORATION FINANCE AND THE THEORY OF INVESTMENT

By Franco Modigliani and Merton H. Miller*

What is the "cost of capital" to a firm in a world in which funds are used to acquire assets whose yields are uncertain; and in which capital can be obtained by many different media, ranging from pure debt instruments, representing money-fixed claims, to pure equity issues, giving

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Law of One Price

- By the Law of One Price the combined values of debt and equity must be the same as the value of the unlevered firm/unlevered equity:
- If the equity were overpriced, investors would sell their stocks, take up a loan and invest into another/similar unlevered stream of cash flows (firm) that is priced correctly.
- If the equity were underpriced, investors would buy all bonds, and sell the unlevered cash flow for the correct price.

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Financing a Firm with Debt and Equity

- Because the cash flows of the debt and equity sum to the cash flows of the project, by the Law of One Price the combined values of debt and equity must be $1000.
  - Therefore, if the value of the debt is $500, the value of the levered equity must be $500.
    - E = $1000 – $500 = $500.

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Example

	Unlevered Firm = Unlevered Equity		*Levered Equity*		*Debt*	
Date 0 Initial value	1000		500		500	
Date 1 – Strong Cash Flow/Return	1400	40%	875		525	5%
Date 1 – Weak Cash Flow /Return	900	-10%	375		525	5%
Date 1 – Expected Cash Flow /Return	1150	15%	625		525	5%

Table 1

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Financing a Firm with Debt and Equity (cont'd)

- Because the cash flows of levered equity are smaller than those of unlevered equity, levered equity will sell for a lower price ($500 versus $1000).
  - However, you are not worse off. You will still raise a total of $1000 by issuing both debt and levered equity.
  - Consequently, you would be indifferent between these two choices for the firm's capital structure.

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Example

	Unlevered Firm = Unlevered Equity		*Levered Equity*		*Debt*	
Date 0 Initial value	1000		500		500	
Date 1 – Strong Cash Flow/Return	1400	40%	875	75%	525	5%
Date 1 – Weak Cash Flow /Return	900	-10%	375	-25%	525	5%
Date 1 – Expected Cash Flow /Return	1150	15%	625	25%	525	5%

Table 1

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The Effect of Leverage on Risk and Return

- Leverage increases the risk of the equity of a firm.
  - Therefore, it is inappropriate to discount the cash flows of levered equity at the same discount rate

- The returns to equity holders are very different with and without leverage.
  - Unlevered equity has a return of either 40% or –10%, for an expected return of 15%.
  - Levered equity has higher risk, with a return of either 75% or –25%.
    - To compensate for this risk, levered equity holders receive a higher expected return of 25%.

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The Effect of Leverage on Risk and Return

- Note that the returns to equity holders are very different with and without leverage.
- Leverage increases the risk of the equity, and may increase the risk of debt.

Debt Value	Equity Value	Unlevered Flow		Equity			Debt		
		Strong	Weak	Strong	Weak	Exp %	Strong	Weak	Exp %
100	900	1400	900	1295	795	16.11 %	105	105	5 %
300	700	1400	900	1085	585	19.29 %	315	315	5 %
500	500	1400	900	875	375	25.00 %	525	525	5 %
800	200	1400	900	560	60	55.00 %	840	840	5 %
≈857.14	≈142.86	1400	900	500	0	75.00 %	900	900	5 %
900	100	1400	900	350	0	75.00 %	1050	900	8.33 %
950	50	1400	900	175	0	75.00 %	1225	900	11.84 %
990	10	1400	900	35	0	75.00 %	1365	900	14.39 %

Table 2

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The link with Beta

- Assume that the returns behave as follows:

	Return Unlevered Equity	Return Market	Risk free
Date 1 (weak economy)	40 %	26 %	5 %
Date 1 (strong economy)	-10 %	-4 %	5 %
Date 1 (expected)	15 %	11 %	5 %

Table 3

- This implies risk Premium on Market = 11%-5%=6%.
- Covariance between market returns and unlevered equity: 0.0375
- Variance of market returns: 0.0225
- Beta of unlevered equity:

$$\beta_{\text{unlevered}} = \frac{0.0375}{0.0225} = 5/3$$

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Relationship between Required Return on Levered and Unlevered equity

- Investors in levered equity will require a higher expected return to compensate for the increased risk.
- M&M's first proposition can be used to derive an explicit relationship between leverage and the equity cost of capital.
- Let us apply the following notation:
  - E - Market value of equity in a levered firm.
  - D - Market value of debt in a levered firm.
  - U - Market value of equity in an unlevered firm.
  - A - Market value of the firm's assets.

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Relationship between Required Return on Levered and Unlevered equity

- **M&M Proposition I** states that: $E + D = U = A \;\; or \;\; V_L = V_U$
- *The total market value of the firm's securities is equal to the market value of its assets, whether the firm is unlevered or levered.*
- **The return on unlevered equity (r_U) :**

$$\frac{E}{E+D} \cdot r_E + \frac{D}{E+D} r_D = r_U$$

is related to the returns of levered equity (r_E) and debt (r_D):

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Relationship between Required Return on Levered and Unlevered equity

- Solving for the return on levered equity:

$$r_E = \underbrace{r_U}_{\text{Risk without leverage}} + \underbrace{\frac{D}{E}(r_U - r_D)}_{\text{Additional risk due to leverage}}$$

- The levered equity return equals the unlevered return, plus a premium due to leverage.

- Using the numbers from Table 2 in the slides

$$15\,\% + (15\,\% - 5\,\%) \cdot \frac{500}{500} = 25\,\%$$

$$15\,\% + (15\,\% - 11.84\,\%) \cdot \frac{950}{50} = 75\,\%$$

- The amount of the premium depends on the amount of leverage, measured by the firm's market value debt-equity ratio, D/E.

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Modigliani and Miller's Proposition II:

- The cost of capital of levered equity is equal to the cost of capital of unlevered equity plus a premium that is proportional* to the market value debt-equity ratio.

$$r_E = r_U + \frac{D}{E}(r_U - r_D)$$

**Be careful: If the required return on debt changes with leverage then the relationship between D/E and required return on levered equity becomes non-linear.*

Capital Budgeting and the Weighted Average Cost of Capital

- If a firm is unlevered, all of the free cash flows generated by its assets are paid out to its equity holders.
- The market value, risk, and cost of capital for the firm's assets and its equity coincide and, therefore: $r_U = r_A$
- If a firm is levered, project r_A is equal to the firm's weighted average cost of capital, or *Unlevered Cost of Capital (pretax WACC)*

$$r_{wacc} \equiv \begin{pmatrix}\text{Fraction of Firm Value}\\ \text{Financed by Equity}\end{pmatrix}\begin{pmatrix}\text{Equity}\\ \text{Cost of Capital}\end{pmatrix} + \begin{pmatrix}\text{Fraction of Firm Value}\\ \text{Financed by Debt}\end{pmatrix}\begin{pmatrix}\text{Debt}\\ \text{Cost of Capital}\end{pmatrix}$$

$$= \frac{E}{E+D} r_E + \frac{D}{E+D} r_D$$

$$r_{wacc} = r_U = r_A$$

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Capital Budgeting and the Weighted Average Cost of Capital

- With perfect capital markets, a firm's WACC is independent of its capital structure and is equal to its equity cost of capital if it is unlevered, which matches the cost of capital of its assets.

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Levered and Unlevered Betas

- Unlevered Beta
  - A measure of the risk of a firm as if it did not have leverage, which is equivalent to the beta of the firm's assets.
- If you are trying to estimate the unlevered beta for an investment project, you should base your estimate on the unlevered betas of firms with comparable investments.

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Levered and Unlevered Betas

- The effect of leverage on the risk of a firm's securities can also be expressed in terms of beta:

$$\beta_U = \frac{E}{E + D}\beta_E + \frac{D}{E + D}\beta_D$$

- Or if we rearrange

$$\beta_E = \beta_U + \frac{D}{E}(\beta_U - \beta_D)$$

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Conclusions

- In a perfect capital markets financial transactions neither add nor destroy value.
- They only represent a repackaging of risk (and therefore return).
- If firms in the real world search for an optimal capital structure then it is because of imperfections and taxes.

Market Imperfections (Chapter 15 Taxes)

- A perfect capital markets exists under:
  - Investors and firms can trade the same set of securities at competitive market prices equal to the present value of their future cash flows.
  - There are no taxes, transaction costs, or issuance costs associated with security trading.
  - A firm's financing decisions do not change the cash flows generated by its investments, nor do they reveal new information about them.
- If capital structure does matter, then it must stem from market imperfections.

Chapter 15: Debt and Taxes

- 15.1 The Interest Tax Deduction
- 15.2 Valuing the Interest Tax Shield
- 15.3 Re-capitalizing to Capture the Tax Shield
- 15.4 Personal Taxes
- 15.5 Optimal Capital Structure with Taxes

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Capital Structure when Considering Corporate Taxation

- Corporations pay taxes on their profits after interest payments are deducted.
- Thus, interest expenses reduce the amount of corporate taxes.
- This creates an incentive to use debt.
- Let us have a look at the example above, and assume: D=500

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Example

	With Leverage		Without Leverage	
	Strong Economy	Weak Economy	Strong Economy	Weak Economy
EBIT	1400	900	1400	900
Interest	25	25	0	0
Net Income before taxes	1375	875	1400	900
Taxes (25 %)	343.75	218.75	350.00	225.00
Flow to Debt	525.00	525.00	0	0
Flow to Equity	531.25	156.25	1050.00	675.00
Flow to Capital	**1056.25**	**681.25**	**1050.00**	**675.00**
Expected	**868.75**		**862.5**	

Table 4

Interest Tax Shield

- Interest Tax Shield: The reduction in taxes paid due to the tax deductibility of interest.

$$\text{Interest Tax Shield} = \text{Corporate Tax Rate} \times \text{Interest Payments}$$

- The cash flows a levered firm pays to investors will be higher than they would be without leverage by the amount of the interest tax shield.

$$\begin{pmatrix}\text{Cash Flows to Investors}\\ \text{with Leverage}\end{pmatrix} = \begin{pmatrix}\text{Cash Flows to Investors}\\ \text{without Leverage}\end{pmatrix} + (\text{Interest Tax Shield})$$

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MM Proposition I with taxes

- The total value of the levered firm exceeds the value of the firm without leverage due to the present value of the tax savings from debt:

$$V^L = V^U + PV(\text{Interest Tax Shield})$$

Implication: The value of the interest tax shield can be found by comparing the value of the levered firm, V^L, to the unlevered value, V^U, of the free cash flow discounted at the firm's unlevered cost of capital, the pretax WACC

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The Weighted Average Cost of Capital with Taxes

- With tax-deductible interest, the effective after-tax borrowing rate is $r(1 - \tau_c)$ and the **weighted average cost of capital** becomes

$$r_{Wacc} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D (1 - \tau_c)$$

$$r_{Wacc} = \underbrace{\frac{E}{E+D} r_E + \frac{D}{E+D} r_D}_{\text{Pretax WACC}} - \underbrace{\frac{D}{E+D} r_D \tau_c}_{\text{Reduction Due to Interest Tax Shield}}$$

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Example

- **Problem**
  - Harris Solutions expects to have free cash flow in the coming year of $1.75 million, and its free cash flow is expected to grow at a rate of 3.5% per year thereafter.
  - Harris Solutions has an equity cost of capital of 12% and a debt cost of capital of 7%, and it pays a corporate tax rate of 40%. If Harris Solutions maintains a debt-equity ratio of 2.5, what is the value of its interest tax shield?

Solution

$$\text{Pretax WACC} = \frac{\text{E}}{E+D} r_E + \frac{\text{D}}{E+D} r_D$$

$$= \left(\frac{1}{1+2.5}\right)12\% + \left(\frac{2.5}{1+2.5}\right)7\% = 8.43\%$$

- Because Harris Solution's free cash flow is expected to grow at a constant rate, have

$$V^U = \frac{\$1.75 \text{ million}}{8.43\% - 3.50\%} = \$35.50 \text{ million}$$

- To compute Harris Solution's levered value, we calculate its WACC:

$$\text{WACC} = \frac{\text{E}}{E+D} r_E + \frac{\text{D}}{E+D} r_D (1-\tau_C)$$

$$= \left(\frac{1}{1+2.5}\right)12\% + \left(\frac{2.5}{1+2.5}\right)7\%(1-0.40) = 6.43\%$$

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Solution cont.

- Thus, Harris Solution's value including the interest tax shield is

$$V^L = \frac{\$1.75 \text{ million}}{6.43\% - 3.50\%} = \$59.73 \text{ million}$$

The value of the interest tax shield is therefore:

$$PV(\text{Interest Tax Shield}) = V^L - V^U = \$59.73 - \$35.50 = \$24.23 \text{ million}$$

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MM Proposition II with taxes and target debt-to-equity ratio

- How does it look?
- It depends on the assumption about the risk of the tax shield
- Lets assume that **the tax shield risk is the same as the risk of the unlevered business**. Then:

$$r_E = r_U + \frac{D}{E}(r_U - r_D)$$

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MM Proposition II with taxes and target debt-to-equity ratio

From M&M I: $V^U + PV(TS) = V^L = E + D$

The same cashflows must have the same expected returns

$$r_U V^U + PV(TS) r_U = E r_E + D r_D$$

$$r_U (E + D - PV(TS)) + PV(TS) r_U = E r_E + D r_D$$

$$r_U (E + D) = E r_E + D r_D$$

$$r_E = r_U + \frac{D}{E}(r_U - r_D)$$

In this case we have $r_U = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D$; $r_U \neq r_{wacc}$

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The Interest Tax Shield with Permanent Debt

- Typically, the level of future interest payments is uncertain due to changes in the marginal tax rate, the amount of debt outstanding, the interest rate on that debt, and the risk of the firm.
  - Here we will consider a special case in which the above variables are kept constant.

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The Interest Tax Shield with Permanent Debt

- Suppose a firm borrows debt *D* and keeps the debt permanently. If the firm's marginal tax rate is τ_c, and if the debt is riskless with a risk-free interest rate r_f, then the interest tax shield each year is $\tau_c \times r_f \times D$, and the tax shield can be valued as a perpetuity:

$$PV(\text{Interest Tax Shield}) = \frac{\tau_c \times \text{Interest}}{r_f} = \frac{\tau_c \times (r_f \times D)}{r_f}$$
$$= \tau_c \times D$$

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The Interest Tax Shield with Permanent Debt

- If the debt is fairly priced, no arbitrage implies that its market value must equal the present value of the future interest payments:

$$\text{Market Value of Debt} = D = PV(\text{Future Interest Payments})$$

$$\begin{aligned} PV(\text{Interest Tax Shield}) &= PV(\tau_c \times \text{Future Interest Payments}) \\ &= \tau_c \times PV(\text{Future Interest Payments}) \\ &= \tau_c \times D \end{aligned}$$

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MM Proposition II with taxes and constant debt

- The cost of capital of levered equity is equal to

$$r_E = r_U + \frac{D}{E}(r_U - r_D)(1 - \tau_c)$$

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MM Proposition II with taxes and constant debt: derivation

From M&M I: $V^U + D\tau_c = V^L = E + D$ (with permanent debt)

The same cashflows must have the same expected returns

$$r_U V^U + D\tau_c r_D = E r_E + D r_D$$

$$r_U V^U = E r_E + D r_D\,(1 - \tau_c)$$

$$r_U (E + D - D\tau_c) = E r_E + D r_D\,(1 - \tau_c)$$

$$E r_E = E r_U + D r_U - D\tau_c r_U - D r_D (1 - \tau_c)$$

$$r_E = r_U + \frac{D}{E}(r_U - r_D)\,(1 - \tau_c)$$

Note that it also implies that $r_U = \frac{E}{E+D-D\tau_c} r_E + \frac{D}{E+D-D\tau_c} r_D (1 - \tau_c)$; $\boldsymbol{r_U \neq r_{wacc}}$

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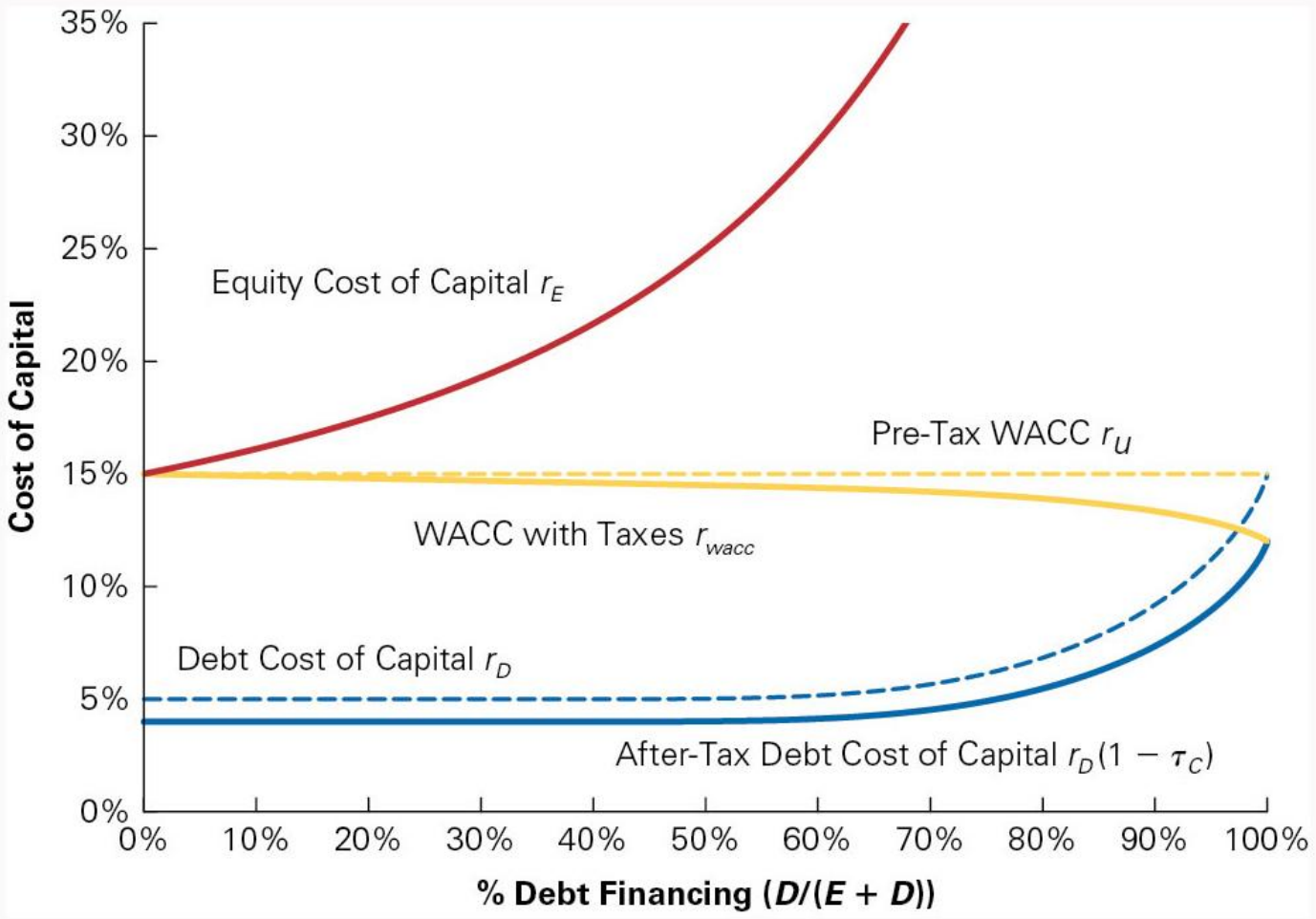
Figure 15.2 The WACC with and Without Corporate Taxes



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Personal Taxes (for self-study)

- The amount of money that an investor want to pay for a security depends on the benefits that the investor will receive after all taxes have been paid.
- Hence, we have to take into account both corporate and private taxation.
- Often there is a difference in the taxation of income from dividends (equity) and interest (debt).

Optimal Capital Structure with Taxes

- Do Firms Prefer Debt?
  - The use of debt varies greatly by industry
  - Firms in growth industries like biotechnology or high technology carry very little debt, while airlines, automakers, utilities, and financial firms have high leverage ratios.
  - Debt ratios also varies among countries
  - A general finding, however, is that firms have lower debt ratios than its optimal regarding the value of the tax shield.
  - A key item missing from the analysis thus far is that increasing the level of debt increases the probability of bankruptcy.

Question form last time

- How might the project's cost of capital change if the firm uses leverage to finance the project?
- Perfect Capital Markets
  - In perfect capital markets, choice of financing does not affect cost of capital or project NPV
- Taxes – A Big Imperfection
  - When interest payments on debt are tax deductible
  - We use WACC

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Summary - No taxes

- MM Proposition I: $V^U = V^L = E + D$
- Weighted average cost of capital without taxes

$$r_{Wacc} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D = r_A = r_U$$

- Return on levered equity MM Proposition II

$$r_E = r_U + \frac{D}{E}(r_U - r_D)$$

- Relationship between levered and unlevered betas

$$\beta_U = \frac{E}{E+D}\beta_E + \frac{D}{E+D}\beta_D$$

$$\beta_E = \beta_U + \frac{D}{E}(\beta_U - \beta_D)$$

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Summary:
Formula Recap – with taxes – and constant debt

With permanent debt

M&M I: $$V^U + D\tau_c = V^L = E + D$$

– The value of the unlevered firm is equal to the value of the levered firm plus the present value of interest tax shield where PV(TS)= $D\tau_c$

– The risk of the tax shield is the same as that of debt, thus, r_D

The same cashflows must have the same expected returns

M&M II: the cost of capital on the unlevered equity in this case is

$$r_E = r_U + \frac{D}{E}(r_U - r_D)\,(1 - \tau_c)$$

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Summary:
Formula Recap – with taxes – and target debt-to-equity rato

With target debt-to-equity ratio

M&M I: $$V^U + PV(TS) = V^L = E + D$$

– The value of the unlevered firm is equal to the value of the levered firm plus the present value of interest tax shield where PV(TS)= $D\tau_c$

– Under the assumption that the risk of the tax shield is the same as that of unlevered equity, thus, r_U

M&M II: the cost of capital on the unlevered equity in this case is

$$r_E = r_U + \frac{D}{E}(r_U - r_D)$$

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